

Research Report: Metal ion diffusion and intercalation in graphene

Keya Joy, EE 499, 2026

Background

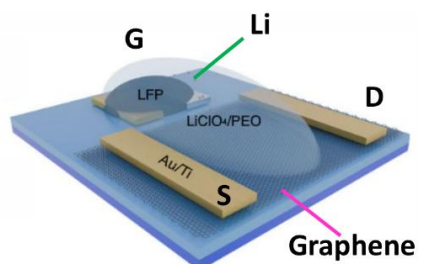


Fig. 1 ECRAM Device

Classic computer architecture has a limited throughput between memory and processing. Inspired by synapses in the human brain, neuromorphic architectures use in-memory computation through artificial synapses that store data as analog conductance. ECRAM uses metal ion movement through a channel to program conductance. With this device, lithium intercalates between graphene layers, changing carrier density and therefore tuning conductance. My research goal is to use ab-initio calculations to further characterize metal ion intercalation and diffusion in layered graphene.

Project A: Energy Barrier

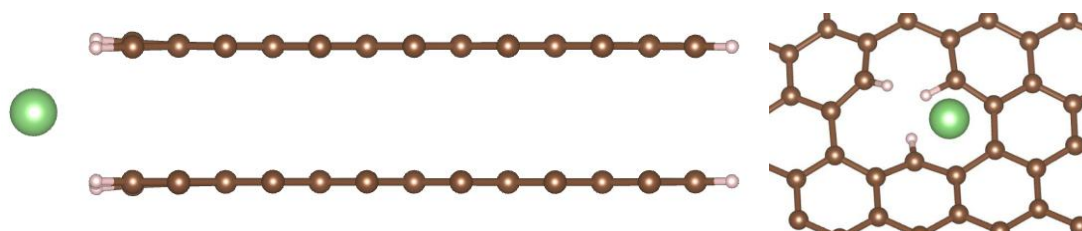


Fig. 2 Lithium entering bilayer graphene from the side (left) and top through monovacancy (right)

Metal ions enter the layered graphene either through the side or from the 'top' through defects known as vacancies. The energy barrier is the energy it takes for atoms to move from one physical configuration to another. This barrier is tied to the threshold voltage needed to intercalate the atoms into the graphene to modulate the conductance and set the value of the device. It requires significantly more energy for intercalation through vacancies, but factors such as metal atom, charge, addition of hydrogen, and size of vacancy can change this energy barrier. I primarily looked at how metals (ie. Li, Cu, Ca), charge on the atom (ie. Li^+ , Cu^+ , Ca^{2+}), and hydrogen termination impacted the barrier. This calculation is done using the Nudged Elastic Band method, which finds the minimum energy pathway between a starting and ending configuration by treating intermediary images like they're connected with springs.



Fig. 3 Visualization of NEB methodology

Setup

Apart from the external variables above, some experiment setup parameters also affect the calculations.

Energy relaxation: A necessary step before NEB calculations is the energy minimization of the first and last image. Early results showed that initial placement of the metal atom in relaxation can cause the atoms to rest in a false-minimum (energy trap), resulting in an unusable energy pathway. If the metal atom is placed directly in the center, it is likely that the energy gradient in all directions is equal, so it seems like it is at a minimum and the relaxation process terminates quickly (fig. 4). After testing several positions with various atoms, I found that if placed anywhere but the center, the relaxed configuration will fall into one of 3 rotationally symmetric placements (fig. 5).

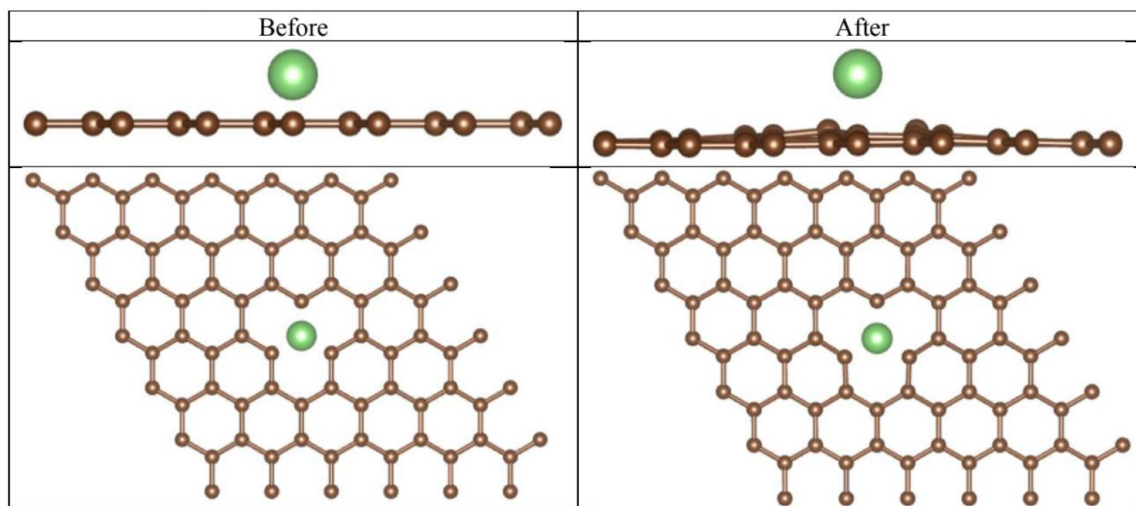


Fig. 4 Lithium before and after when lithium is placed in the center of the vacancy

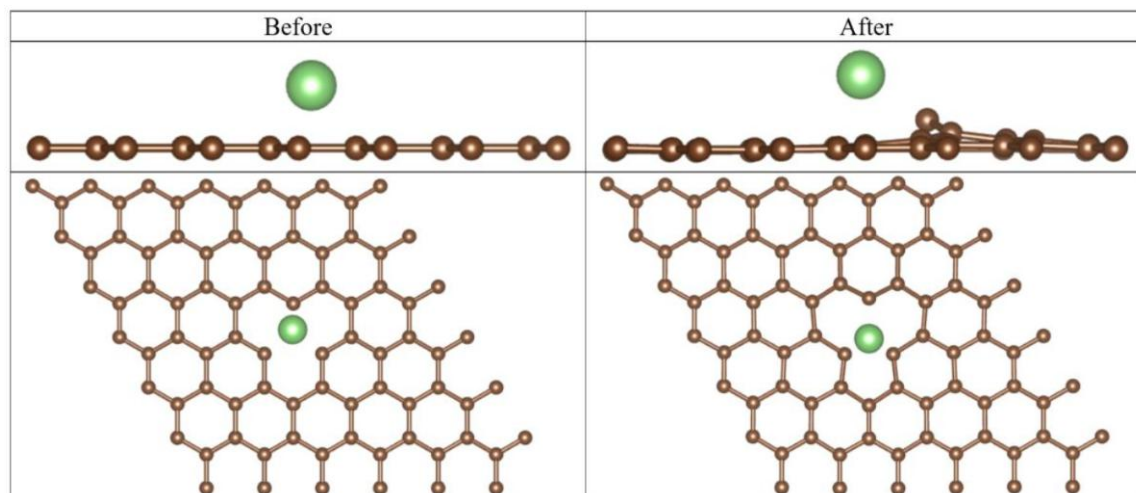


Fig. 5 Lithium before and after relaxation when lithium is placed off-center

The effect of the incorrect minimization is that the NEB algorithm may find the correct energy minimum which would end up being in an intermediary image, skewing the energy barrier (fig. 6). Fig. 7 shows what the pathway looks like when the images are minimized correctly.

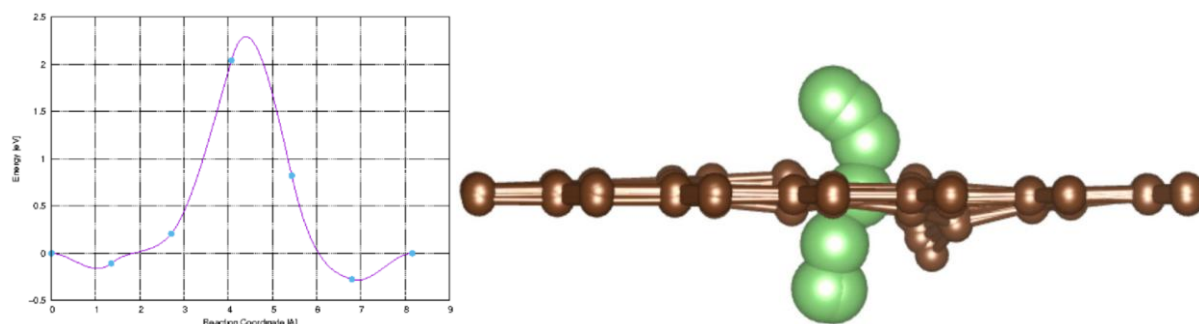


Fig. 6 Pathway when lithium is placed in the center for relaxation

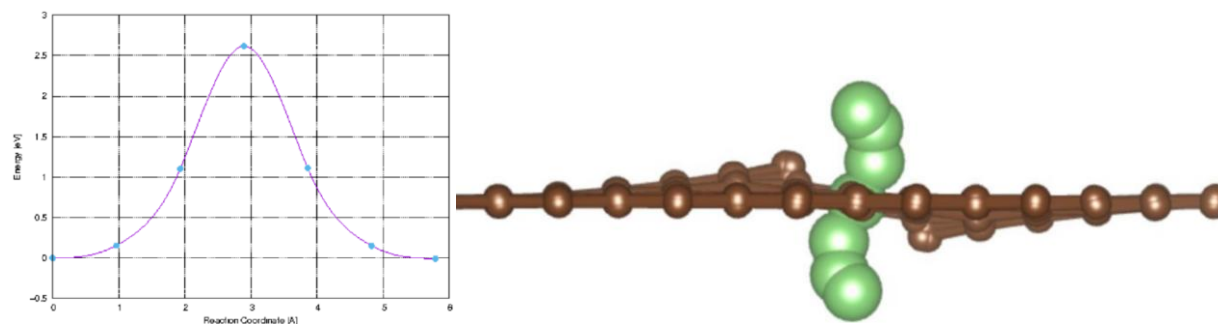


Fig. 7 Pathway when lithium is placed off-center for relaxation

Precision: Precision (PREC) is another parameter in the INCAR file for both the minimization and the NEB, which defines other parameters like ENCUT. I learned that it is often more important to match the precision of the minimization and NEB than to use the best precision possible. For example, when calculating the energy barrier of diffusion of Li in AB-stacking graphene, mismatched precision results in a 0.2 eV barrier, whereas matching precision resulted in a 0.077 eV barrier (approaching predicted value).

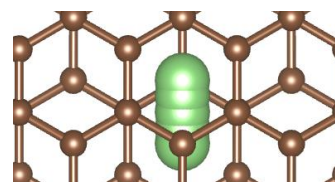


Fig. 8 Lithium diffusion in AB-stacking graphene

Monopole-monopole corrections: monopole-monopole corrections (LMONO) did not have a significant effect on the energy barrier (NEB) or charge (Bader charge analysis). However, it also does not have a significant effect on the length of the computation job either.

Climbing NEB: A version of NEB that forces an image into the transition point. Sometimes even with an odd number of atoms and an assumption of a singular highest energy point, the middle image will not find that point, making it difficult to calculate the energy barrier. C-NEB often

helps with this, by trying to force the symmetrical energy pathway without increasing computation time. This was more helpful with some atoms than others. For example, copper exhibited a symmetric energy pathway without C-NEB, but lithium does not.

Post-Analysis

Bader charge analysis finds the charge on each atom by defining atom boundaries along the points in space where charge density is at a minimum. By integrating the charge density with respect to volume within an atom's boundaries, you can find the charge of the atom. The charge on a metal ion indicates how much charge the atom 'donates' to the graphene which is what changes the conductivity of the material (data being stored).

Results

Energy barrier (eV) scenarios	Li	Ca	Mg	Cu	Fe
Atom, no charge, no H (CNEB)	2.619	5.608	2.744	1.113	2.660
Hydrogen termination	1.744	5.330	2.753		
Charge (Li+, Ca2+, Mg2+, Cu+)	1.54	6.675	3.15		
Charge and hydrogen termination	7.8*	11.9*	IP	IP	

*Unfinished, values expected to decrease

Ongoing work: I am running into some errors with the ion + hydrogen termination jobs, namely receiving a warning "Sub-Space-Matrix is not Hermitian in DAV". We suspect this may be due to the middle image (when the ion is in line vertically with the graphene) containing atoms that are too close. This could be fixed by slightly altering the position of the initial minimization.

Project B: Lithium Diffusion

Molecular dynamics studies how a system of atoms move over time. Ab initio MD (AIMD) uses DFT to estimate the forces acting on each atom and find their position after a time step using Newton's law. This part of the project studies lithium (as the metal ion) diffusion once it is already intercalated into the graphene (as compared to Project A, which studies the intercalation). Here, we are trying to calculate the diffusion coefficient of lithium, which measures how fast lithium travels in bilayer graphene in cm^2/s .

Diffusion coefficient calculations

With a cartesian trajectory of atoms the diffusion coefficient can be calculate by look at the mean squared displacement (MSD) for each time. Using the Einstein equation for 3D, the theoretical relation between diffusion coefficient and MSD is shown in (1). The Einstein diffusion relation states that MSD grows linearly with time, so since we're dividing by t , this is akin to finding the slope of the MSD vs. t graph and dividing by 6 because this is in 3D, shown in (2).

$$D = \lim_{t \rightarrow \infty} \frac{MSD(t)}{6t} \quad (1)$$

$$D = \lim_{t \rightarrow \infty} \frac{MSD'(t)}{6} \quad (2)$$

The typical way to calculate MSD is shown in (3), where M is the number of atoms, and $r_i(t)$ is the position of atom i and time t . Essentially each MSD value is the average mean squared displacement of the atoms from time t since the origin point. This works well when you have several atoms, but when you have just a few (in my case $M = 1$), the calculation becomes very noisy.

$$MSD(t) = \frac{1}{M} \sum_{i=1}^M [r_i(t + t_0) - r_i(t_0)]^2 \quad (3)$$

Another way to calculate the MSD with one time is using time-averaged MSD (TAMSD), which I happened to independently re-discover. This method uses a sliding window $[n, t + n]$ and averages the MSD across the window for each time interval t . This is depicted in (4), where T is the number of time steps, $r(t)$ is the position of the atom at time t .

$$MSD_2(t) = \frac{1}{T - t} \sum_{n=1}^{T-t-1} [r(t + n) - r(n)]^2 \quad (4)$$

Finally, a third calculation I came up with is combining the two methods which I find works well when you have a few atoms, where (3) still is noisy. In (5), the displacement is averaged across M atoms for each sliding window of interval t .

$$MSD_3(t) = \frac{1}{T-t} \sum_{n=0}^{T-t-1} \frac{1}{M} \sum_{i=1}^M [r_i(t+n) - r_i(n)]^2 \quad (5)$$

With the following parameters, the MSD vs. time is plotted using MSD (3) and TAMSD (4) in Fig. X and Fig. X respectively. This makes the noise reduction apparent when you only have one lithium.

- 300K
- 7x7 graphene supercell (196 carbon)
- 3 fs timestep
- 3000 steps
- 1 lithium

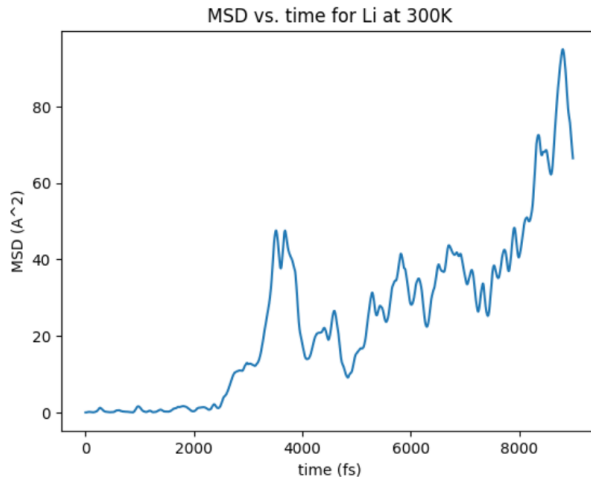


Fig. 9 MSD for 1 lithium

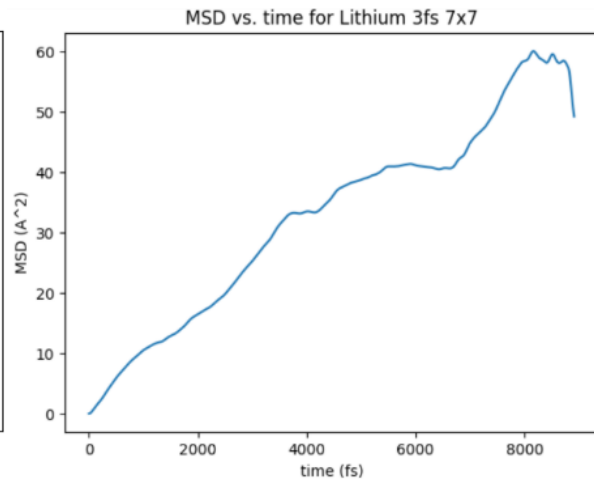


Fig. 10 TAMSD with 1 lithium

With the following parameters, the MSD vs. time is plotted using MSD (3) and the modified MSD (5) in Fig. X and Fig. X respectively. With 4 lithium, the regular MSD is still noisy, so the time-averaging reduces noise.

- 300K
- 6x6 graphene supercell (196 carbon)
- 1 fs timestep
- 3000 steps
- 4 lithium

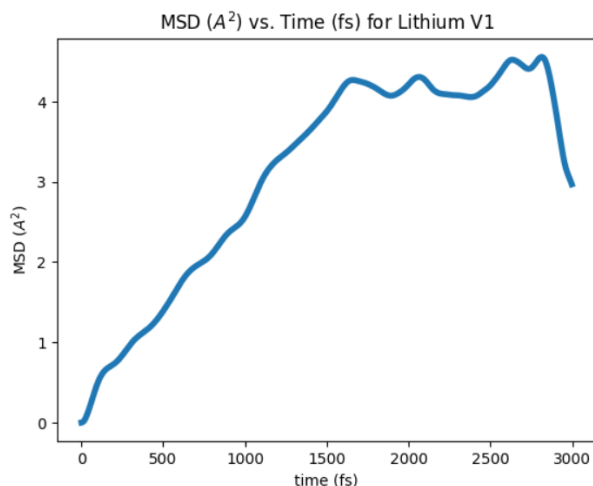


Fig. 11 TAMSD averaged with 4 lithium

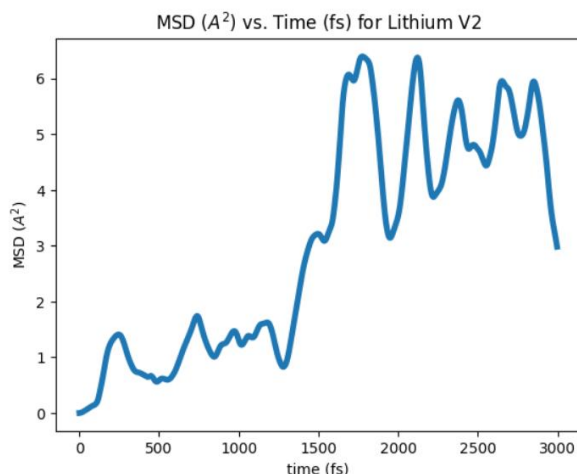


Fig. 12, MSD with 4 lithium

Other Analysis

Sliding: Sliding is a phenomenon I encountered a lot where the two graphene sheets slide horizontally in opposite directions (Fig. 13). I am not sure what causes this but I observed that higher temperature and a small supercell both increase sliding. When I decreased the temperature to 300K and increased the supercell size to 7x7, very drastic sliding stops occurring (Fig. 14, as example with temperature).



Fig. 13 Sliding depiction

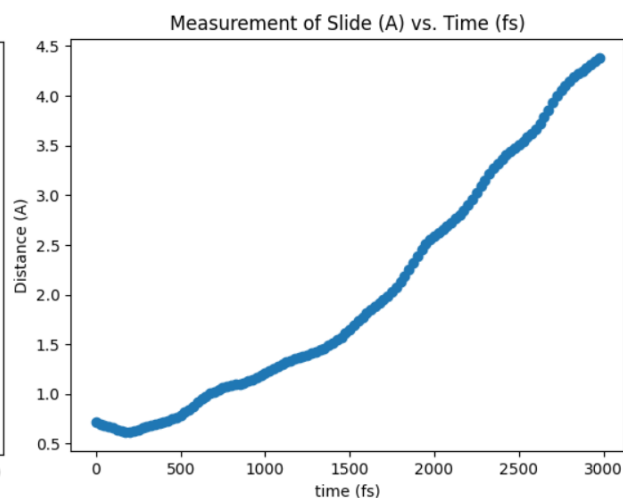
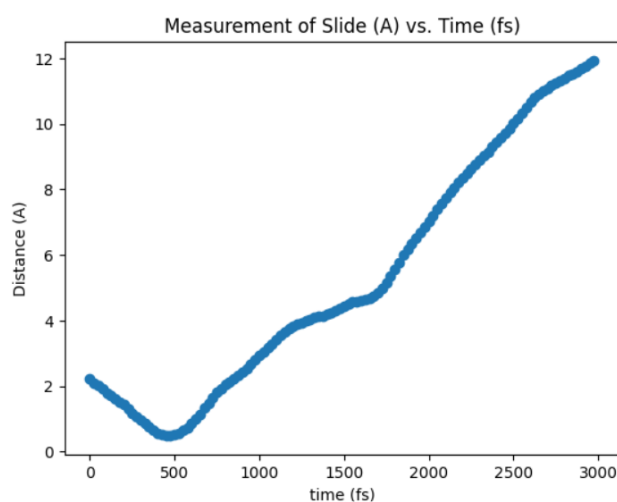


Fig. 14 Slide vs. time at 600K (left) compared to that at 300K (right) for 7x7 supercell with 1 lithium. Note the difference of 8Å of sliding.

I did pose a question on the VASP forum about sliding and fractional coordinates outside the range [0, 1] but I still am not sure the entirety of what affects the sliding.

Interlayer Distance: Interlayer distance is another measurement I looked at over time for jobs and found that it always oscillates between about 3.4 and 4.2 Å (compared to the initial POSCAR value of 3.6 Å). The frequency of the oscillation is around 1.3-2 cycles per ps.

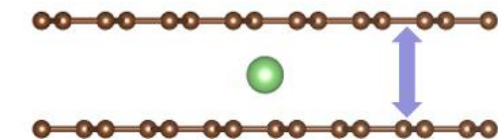


Fig. 15 Interlayer Distance depiction

I also calculated the standard deviation of the interlayer distance, which indicates that the sheets appear to ripple as opposed to moving up and down completely flat (Fig. 16). I don't believe the interlayer distance has an effect on the sliding based on my experiments.

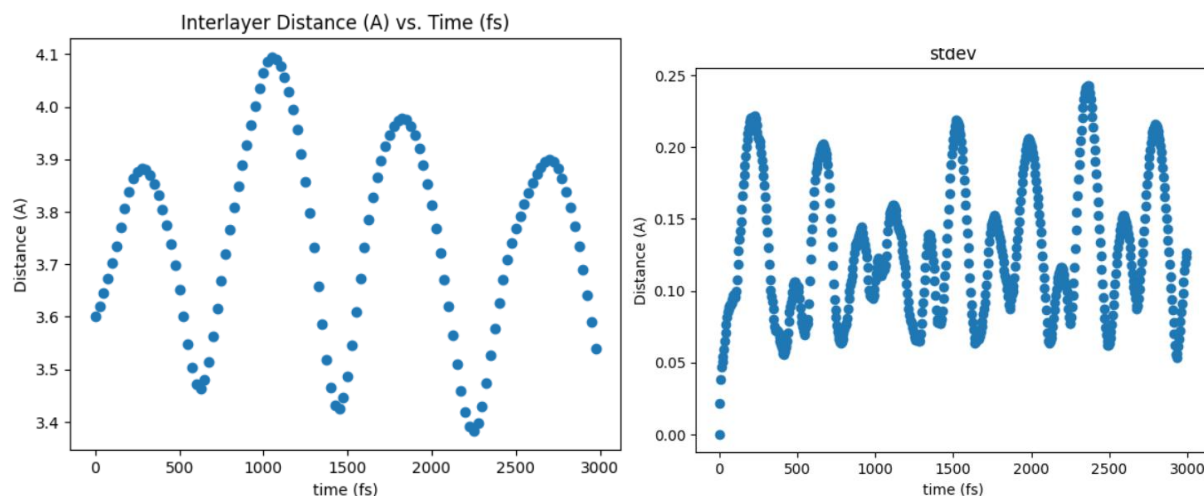


Fig. 16 Interlayer distance vs. time and standard deviation of interlayer distance vs. time for 7x7 bottom layer frozen

Results

The diffusion coefficient of lithium that I calculated is $5.717 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$, which is within but on the higher end of results found in literature (10^{-5} to $7 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$). Through several trials, the coefficients I calculated varied greatly but sometimes examining other characteristics like sliding, interlayer distance, and relative lithium position gave insight into the fluctuations.

Some of the variables I tested included:

- Temperature: 300K vs 600K
 - I started by testing but in 600K, the graphene layers tended to slide apart. Additionally, since 300K was more successful and more relevant to the device, I decided to stick with 300K for future trials
- Time step: 3fs vs. 1fs
 - I started with 3fs to get a longer total time frame (3ps), but after trying a 1fs timestep and getting results, I stuck with that since it is more accepted. Sometimes with a larger timestep the program can overestimate the forces on atoms within that time step.

- Supercell: 5x5, 6x6, 7x7 graphene + lithium primitive cell
 - I began with a 5x5 supercell and 1 lithium at both 300K and 600K but noticed that the sheets would slide apart. Switching to 7x7 (at 600K) still had sliding sheets but 300K at 7x7 supercell slid much less. With and without lithium still lead to sliding issues and I am not sure what causes this result. I decided to stick with 7x7 supercell, as bigger supercells are much slower due to exponential increase in number of atoms.
- POTCAR: pseudopotentials for the Lithium
 - I tried both 1 electron and 3 electron pseudopotentials for lithium. I stuck with 1 electron because it didn't show much effect on my calculations and the 3 electron pseudopotential runs took much longer
- KPOINTS: 1x1x1 vs. 3x3x1
 - For faster calculations I started with gamma point only, but later switching to 3x3x1 sampling to see if this could improve accuracy. In the Z direction there is no need for increased sampling but in the horizontal plane (X and Y), VASP AIMD samples the Brillouin zone 9 times (finer sampling mesh) for DFT calculations.
- Lithium: 1-4 lithium
 - For most jobs I used just 1 lithium but I did simulate 6x6 graphene supercell with 4 lithium (more details below)
- Freezing bottom layer
 - I tried some runs with and without freezing the bottom layer of the bilayer graphene (restricting its movement in the POSCAR file)

The following table shows a summary of lithium diffusion coefficients for different AIMD parameters. Unless specified, the temperature is 300K, the graphene supercell is 7x7, and the coefficient is calculated with 3000 steps of 1 femtosecond each. Additionally, assume 1x1x1 K-point mesh and that all atoms can freely move.

Parameters	Diffusion Coefficient (cm ² /s)
Expected (in literature)	$1 \cdot 10^{-5} - 7 \cdot 10^{-5}$
Timestep = 3 fs	$1.84 \cdot 10^{-4}$
5x5 graphene, bottom layer frozen	$1.10 \cdot 10^{-4}$
Bottom layer frozen	$5.717 \cdot 10^{-5}$
Bottom layer frozen, continued with previous final step	$6.16 \cdot 10^{-6}$
Bottom layer frozen, 3x3x1 K-point mesh	$7.86 \cdot 10^{-5}$
3x3x1 K-point mesh	$8.50 \cdot 10^{-5}$
6x6 graphene with 4 lithium	$6.04 \cdot 10^{-5}$

6x6 Supercell with 4 lithium and hexagon confinement: This cell has a ratio of 36 carbon to 1 lithium and in other papers it has been shown that this concentration can turn AB-stacking graphene into AA-stacking. While I did not see that result, I examined each of the atom's behavior, shown below. By the Y-axis (MSD) ranges, each atom has a different amount of movement, which is also apparent when viewing the trajectory with VMD 2.0. One thing I noticed with this trial and others is that whenever the graphene does skew to AA-stacking (happens when graphene slides so it changes back and forth between AA and AB-stacking), a lithium sometimes gets 'confined' to that carbon hexagon (Fig. 18), resulting in a lower diffusion coefficient. For example, both atoms 1 and 4 have a much smaller maximum MSD of $1.4 \text{ \AA}^2$ and $2.5 \text{ \AA}^2$ respectively, compared to $14 \text{ \AA}^2$ and $8 \text{ \AA}^2$ for atoms 2 and 3.

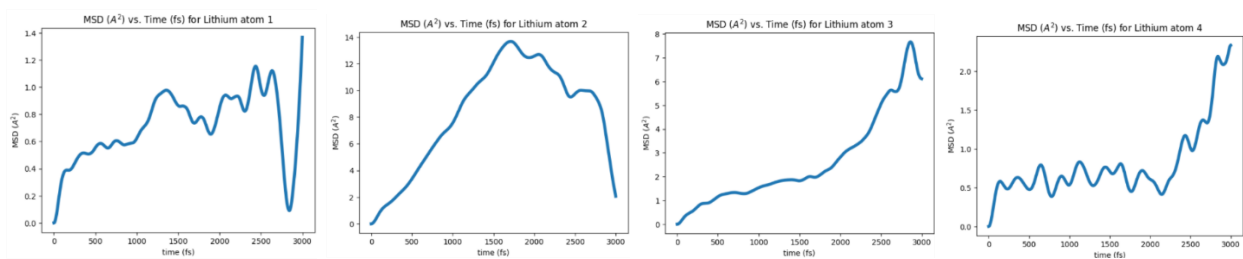


Fig. 17 MSD for each of the 4 lithium atoms

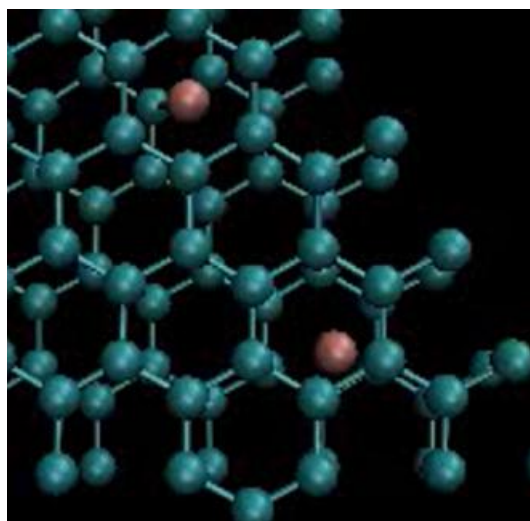


Fig. 18 Snippet from the trajectory where the bottom right atom (atom 4) appears to stay within that AA-stacked hexagon for hundreds of time steps

I believe this could coincide with Simon's finding that the energy barrier is higher for diffusion in AA-stacking (calculated with NEB). Additionally, since I did a 6x6 supercell with 4 lithium as opposed to a 3x3 supercell with 1 lithium, the lithium atoms affect each other more, even though the system is treated as periodic.

Additionally, I tried two methods for calculating the diffusion coefficient: one being the one described in (5), and the other being just averaging the diffusion coefficient of each atom. They provided almost the exact same value; the only difference is probably due to floating point rounding (6.0449733096110804 vs. 6.0449733096110784).

Hexagon confinement: Another example of this phenomenon is with the continuation of the bottom layer frozen MD run. The continuation takes another 3000 steps starting where the first run ended, but the diffusion coefficients are wildly different ($5.717 \cdot 10^{-5}$ vs. $6.16 \cdot 10^{-6} \text{ cm}^2 \text{ s}^{-1}$). This is also apparent in their MSD graphs, where you can see in Fig. 19b between 0 and 2.7 ps, the average MSD stays under $1.5 \text{ \AA}^2$. The trajectory also shows the same phenomenon of hexagon confinement, which isn't apparent in the first trajectory, because the sheets keep sliding so it doesn't get stuck in the AA-stacking.

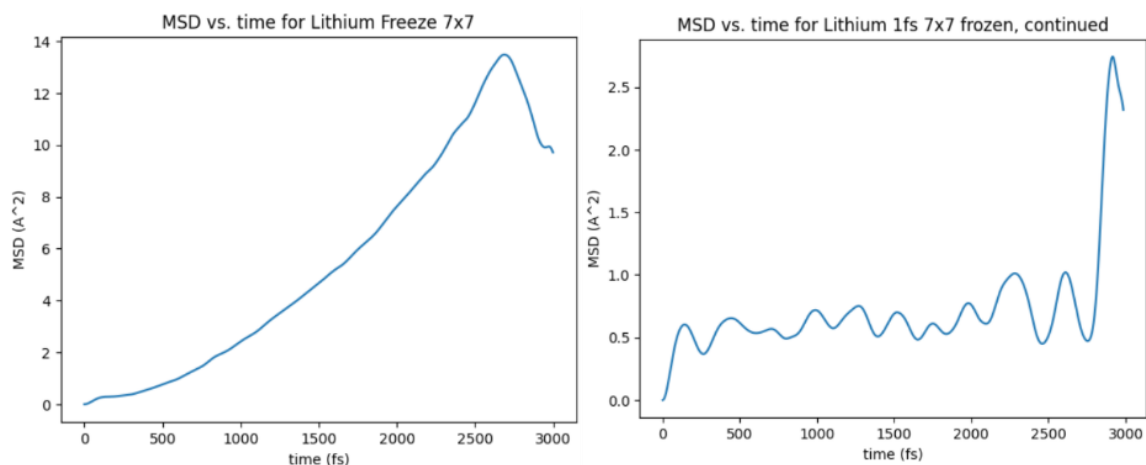


Fig. 19 MSD graphs for 7x7 graphene cell AIMD with one lithium and the bottom layer frozen

Placing Lithium outside unit cell: Frequently the fractional coordinates, whose values are typically in between 0 and 1 (indicating unit cell), will leave those bounds. This indicates that the atom moved relative to the unit cell placement, which is seen a lot with the sliding effect. We also wanted to see what happens if the lithium is placed outside the unit cell to start because in previous runs, once the lithium leaves the unit cell it rarely comes back. However, I discovered that VASP does not let you place atoms outside the unit cell in the initial configuration, and it will wrap that value around. For example if an atom's fractional coordinates are [0.5 1.1 -0.2] it will write it as [0.5 0.1 0.8] when the job begins. This also becomes an issue when I merge two trajectories together (starting one where the previous left off) since in absolute coordinates, it would appear that the lithium is jumping across the cell between the end of the first trajectory and start of the second.

My personal guess is that the location of the lithium does not matter for calculations, and statistically an object in 2D will return to its origin but perhaps the timescale would need to be

Next steps

- 3x3 supercell with 1 lithium to see compare its behavior and trajectory to the 6x6 supercell with 4 lithium
- Placing the lithium close to the boundary of the cell and using a higher K-point density
- Lithium entering Zigzag and Armchair bilayer graphene (elaborated on below)

Fig. 21 Initial configuration of Li entering Zig-zag graphene

Fig. 22 Errors in slurm output file for Zig-zag and Armchair jobs run on HPC cluster Hyak Klone using STF nodes & memory